

IT 123 — Week 6 Lab Report

Network configuration and management • Ubuntu Server only

Student: Not provided **VM:** ubuntu_server **OS:** Ubuntu 26.04.1 LTS **Date:** 27 September 2026

Outcome: The existing VM was changed from DHCP to a working static IPv4 address, tested for routing, internet access, and DNS, then returned to its original DHCP configuration. The final DHCP state was verified.

Network choice

The guide specifies **192.168.1.150/24** and gateway **192.168.1.1**. This VM is connected to a VirtualBox NAT network **10.0.2.0/24**, so those guide values would not reach its actual gateway. The temporary static test used **10.0.2.150/24**, gateway **10.0.2.2**, and DNS **8.8.8.8 / 8.8.4.4**. This tested the same netplan procedure without disconnecting the VM.

Step 1 — Inspect and back up DHCP

```
ip -4 a show dev enp0s3
ip route
resolvectl dns enp0s3
cat /etc/netplan/00-installer-config.yaml
```

The starting address was **10.0.2.15/24** from DHCP; the default route used **10.0.2.2**. The original netplan file was saved and a VirtualBox snapshot named **Week6-before-network** was taken before edits.

```
10.0.2.2 dev enp0s3 proto dhcp scope link src 10.0.2.15 metric 100
192.168.100.1 via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
Link 2 (enp0s3): 192.168.100.1 fd17:625c:f037:2::3
NETPLAN BEFORE
# This is the network config written by 'subiquity'
network:
  ethernets:
    enp0s3:
      dhcp4: true
      dhcp6: true
      match:
        macaddress: 08:00:27:30:e4:a2
      set-name: enp0s3
  version: 2
kadmin@it123:~$ _
```

Figure 1. Original netplan file with DHCP enabled. Full console capture and baseline output are in the repository.

Step 2 — Apply and verify a static address

The saved static file was installed at `/etc/netplan/00-installer-config.yaml`. `sudo netplan generate` checked the file, then `sudo netplan try --timeout 120` applied it with automatic rollback available. The new connection was checked before the trial was accepted.

```
enp0s3:
  dhcp4: false
  addresses: [10.0.2.150/24]
  routes: [{to: default, via: 10.0.2.2}]
  nameservers:
    addresses: [8.8.8.8, 8.8.4.4]
```

```
# Temporary lab configuration for this VM's VirtualBox NAT network.
# The guide's 192.168.1.150/24 requires a different virtual network.
network:
  version: 2
  ethernets:
    enp0s3:
      dhcp4: false
      dhcp6: true
      match:
        macaddress: 08:00:27:30:e4:a2
      set-name: enp0s3
      addresses:
        - 10.0.2.150/24
      routes:
        - to: default
          via: 10.0.2.2
      nameservers:
        addresses: [8.8.8.8, 8.8.4.4]
enp0s3          UP          10.0.2.150/24
kadmin@it123:~$ _
```

Figure 2. Temporary static netplan configuration and active 10.0.2.150/24 address.

Step 3 — Test the static connection

```
ping -c 4 10.0.2.2
ping -c 4 8.8.8.8
ping -4 -c 4 google.com
nslookup google.com
traceroute -4 -m 12 -q 1 -w 1 google.com
ss -tuln
```

Check	Observed result
Gateway / public IP / domain ping	4 replies out of 4 for each target; 0% loss
DNS lookup	google.com resolved to IPv4 and IPv6 addresses
Traceroute	Hop 1 was 10.0.2.2; hops 2–12 did not answer
Listening sockets	SSH on port 22 and local DNS listeners were present

Static connection evidence

The internet and name pings succeeded even though intermediate traceroute probes did not receive replies. Asterisks in traceroute mean no response to those specific probes; they do not establish that ordinary traffic failed.

```
GATEWAY PING
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=64 time=1.23 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=64 time=1.06 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=64 time=0.837 ms
64 bytes from 10.0.2.2: icmp_seq=4 ttl=64 time=1.23 ms

--- 10.0.2.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3026ms
rtt min/avg/max/mdev = 0.837/1.089/1.229/0.160 ms
INTERNET PING
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=15.7 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=15.8 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=64 time=56.4 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=64 time=159 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 200505ms
rtt min/avg/max/mdev = 15.734/61.807/159.302/58.675 ms
NAME PING
PING google.com (172.217.27.78) 56(84) bytes of data.
64 bytes from nrt12s15-in-f14.1e100.net (172.217.27.78): icmp_seq=1 ttl=64 time=126 ms
64 bytes from nrt12s15-in-f14.1e100.net (172.217.27.78): icmp_seq=2 ttl=64 time=24.8 ms
64 bytes from nrt12s15-in-f14.1e100.net (172.217.27.78): icmp_seq=3 ttl=64 time=17.3 ms
64 bytes from nrt12s15-in-f14.1e100.net (172.217.27.78): icmp_seq=4 ttl=64 time=19.9 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3311ms
rtt min/avg/max/mdev = 17.332/46.910/125.626/45.526 ms
kadmin@it123:~$
```

Figure 3. Gateway, public-IP, and domain-name ping results from the static configuration.

```
DNS LOOKUP
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 172.217.27.78
Name:   google.com
Address: 2404:6800:4017:80a::200e

TRACEROUTE
traceroute to google.com (172.217.27.78), 12 hops max, 60 byte packets
 1 _gateway (10.0.2.2)  5.298 ms
 2 *
 3 *
 4 *
 5 *
 6 *
 7 *
 8 *
 9 *
10 *
11 *
12 *
kadmin@it123:~$ _
```

Figure 4. DNS lookup and traceroute from the static configuration. The first hop is the VirtualBox NAT gateway.

Step 4 — Restore DHCP and check the final state

The saved original file was put back, syntax-checked, and applied using `sudo netplan try --timeout 120`. Before acceptance, the VM showed the DHCP address **10.0.2.15/24**, default gateway **10.0.2.2**, successful public-IP ping, and successful DNS resolution. The active file matched the original backup exactly.

```
# This is the network config written by 'subiquity'
network:
  ethernets:
    enp0s3:
      dhcp4: true
      dhcp6: true
      match:
        macaddress: 08:00:27:30:e4:a2
      set-name: enp0s3
  version: 2
enp0s3 UP 10.0.2.15/24 metric 100
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
10.0.2.2 dev enp0s3 proto dhcp scope link src 10.0.2.15 metric 100
192.168.100.1 via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
kadmin@it123:~$ _
```

Figure 5. Restored original DHCP netplan file, active address, and default route.

Final verification	Observed result
Gateway 10.0.2.2	4 of 4 replies
Internet 8.8.8.8	4 of 4 replies
google.com ping	4 of 4 replies
nslookup google.com	IPv4 and IPv6 addresses returned

Scope and evidence

Only one Ubuntu VM was available. It demonstrated the static and DHCP states sequentially; a simultaneous Server A-to-Server B ping was **not performed**. A second VM on a shared network is required for that part of the exercise. Optional Windows Server work was excluded by request. Temporary host-side SSH forwarding rules used during the lab were removed.

Repository folder **Week6/** contains this report, the full unedited text outputs in **evidence/**, the complete screenshots in **screenshots/**, the tested static netplan file in **scripts/**, and a step-by-step explanation in **lab_notes.md**.